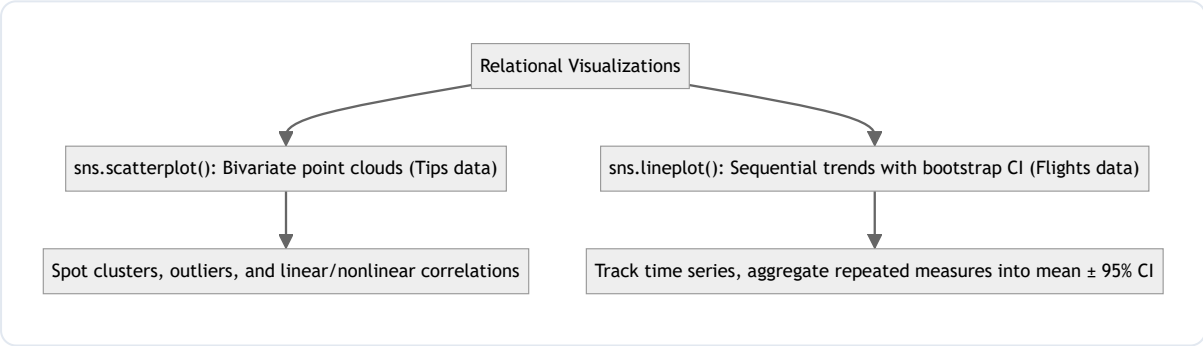
Lecture 19.12: Relational Plots in Seaborn (`scatterplot` & `lineplot`)

📌 Topic Overview

In this lecture, Shraddha ma'am explains the two specialized **axes-level relational plot functions**: `sns.scatterplot()` for observing individual bivariate point clouds and `sns.lineplot()` for tracking continuous temporal trends with automatic statistical aggregation and confidence interval shading.

🏛️ Axes-Level Relational Functions



📐 Mathematical Formulation: Bootstrap Confidence Intervals

When multiple measurements exist for each time point t (e.g. 12 months for each year in the flights dataset), `sns.lineplot()` computes the sample mean:

$$\bar{y}_t = \frac{1}{M} \sum_{m=1}^{M} y_{t,m}$$

The translucent shaded band represents the **95% Bootstrap Confidence Interval** around the mean, calculated by resampling monthly observations with replacement to estimate sampling variance without normality assumptions.

💻 Python Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

def plot_scatter_and_line_relationships(show_plot=True):
    """
    Demonstrates Axes-level relational plots in Seaborn:
    1. sns.scatterplot() for bivariate continuous relationships (total_bill vs tip
    2. sns.lineplot() for temporal trends with automatic aggregation (flights data

    Returns:
        dict: Summary metrics for both plots.
    """
    sns.set_theme(style="whitegrid")
    tips = sns.load_dataset("tips")
    flights = sns.load_dataset("flights")

    fig, axes = plt.subplots(1, 2, figsize=(14, 5))

    # 1. Scatter Plot on axes[0]
    sns.scatterplot(
        data=tips,
        x="total_bill",
        y="tip",
        hue="time",
        palette="deep",
        ax=axes[0]
    )
    axes[0].set_title("Scatter Plot: Total Bill vs Tip (by Time)", fontweight="bol

    # 2. Line Plot on axes[1] (Aggregates multiple months per year into mean + 95%
    sns.lineplot(
        data=flights,
        x="year",
        y="passengers",
        color="crimson",
        marker="o",
        ax=axes[1]
    )
    axes[1].set_title("Line Plot: Airline Passengers Over Years (Mean + CI)", font

    plt.tight_layout()
    if show_plot:
        plt.show()

    return {
        "tips_count": len(tips),
        "flights_year_range": (int(flights["year"].min()), int(flights["year"].max
    }
```

```
# --- Execution & Verification ---
info = plot_scatter_and_line_relationships(show_plot=False)
print(f"Relational Plots Metadata: {info}")
```

🔑 Key Takeaways

1. `sns.scatterplot()` and `sns.lineplot()` accept an `ax=axes[i]` argument, integrating directly into Matplotlib multi-panel subplots.
2. `sns.lineplot()` automatically computes the mean and shades the 95% confidence interval across repeated measurements.
3. Pass `errorbar=None` in `lineplot` if you wish to disable the translucent confidence interval band.